

Review on Functions and Function Notation (Homework)

① If $f(x) = 4 - x^2$, find i) $f(1)$, ii) $f(2)$, and iii) $f(-3)$.

② If $f(x) = \sqrt{3x-1}$, find i) $f(10)$, ii) $f(1)$, and iii) $f(3)$.

③ If $f(x) = 2^{x+4}$, find i) $f(0)$, ii) $f(-1)$, and iii) $f(-4)$.

④ If $f(x) = |x+5| - 2$, find i) $f(6)$, ii) $f(0)$, and iii) $f(5)$.

⑤ If $f(x) = \frac{3}{x^4}$, find i) $f(2)$, ii) $f(1)$, and iii) $f(0)$.

⑥ If $f(x) = 2 + x$, fill in the table below.

x	f(x)
0	
-6	
π	
$\frac{3}{4}$	

⑦ If $f(x) = 10 + \sqrt{x}$, fill in the table below.

x	9	36	18	-4
f(x)				

→ Hint: Functions do not have imaginary outputs.

⑧ If $f(x) = \frac{2}{x-1} + x^3 + 6$, fill in the table below.

x	f(x)
0	
5	

⑨ If $f(x) = 2x^2 - x + 7$, fill in the table below.

x	-1	3	2	-10
f(x)				

⑩ If $f(x) = \frac{1}{4x}$, find i) $f(b)$, ii) $f(0.2)$, and iii) $f(\frac{1}{4})$.

⑪ If $f(x) = -10x^2 + 5x$, find i) $f(a+1)$, ii) $f(\pi)$, and iii) $f(c)$.

⑫ If $f(x) = x - 100 - \sqrt{3x+2}$, find i) $f(2.5)$, ii) $f(3^{b+1})$, and iii) $f(0)$.

In the tables below, x represents the inputs and y represents the outputs. Circle the tables that could not represent functions.

⑬

x	y
3	9
2	8
1	7
0	6
-1	5
-2	4

x	y
0	-1
1	10
7	5
2	-5
3	4
1	9

x	y
11	14
0	17
8	19
3	7
20	21
6	13

x	y
100	25
200	50
300	75
400	100
500	125
600	150

⑭

x	-6	4	3	-8	5	6
y	8	4	5	6	7	9

x	9	10	4	3	-1	2
y	3	5	6	2	0	1

x	14	12	20	14	30	32
y	13	20	11	15	16	0

x	13	15	10	11	17	17
y	0	1	2	3	4	5

(15)

X	$\frac{1}{2}$	0	3	$4\frac{1}{3}$	$2\frac{1}{5}$	5
Y	7	20	4	13	8	6

X	-1	1	-2	-3	7	4
Y	5	5	0	8	10	12

X	Y
14	6
15	7
16	6
17	7
18	6
19	7

X	Y
13	1
20	3
19	5
18	7
21	8
13	9

(16)

X	Y
10	7
3	4
8	19
10	7
10	6
7	8

X	Y
13	15
10	15
18	15
20	15
25	15
26	15

X	Y
8	4
2	6
7	9
6	11
8	13
9	15

X	Y
16	1
17	1
17	2
19	1
20	1
20	-1

(17)

X	3	8	8	6	-2	-1
Y	10	2	2	0	4	5

X	7	4	7	2	7	8
Y	3	5	3	5	3	8

X	10	3	8	10	11	11
Y	4	4	9	4	9	10

X	-4	-2	4	3	-2	7
Y	7	8	9	9	8	9

(18)

X	5	3	0	-1	-1	-1
Y	7	6	2	4	5	4

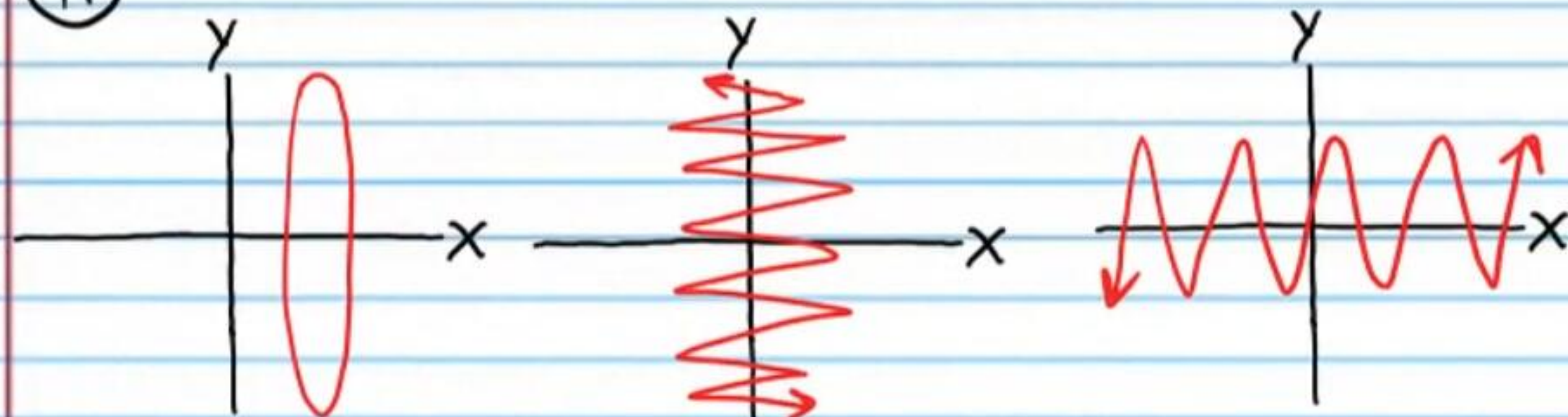
X	7	5	9	7	11	11
Y	7	7	7	4	7	7

X	Y
0	0
1	13
2	15
3	15
1	13
0	100

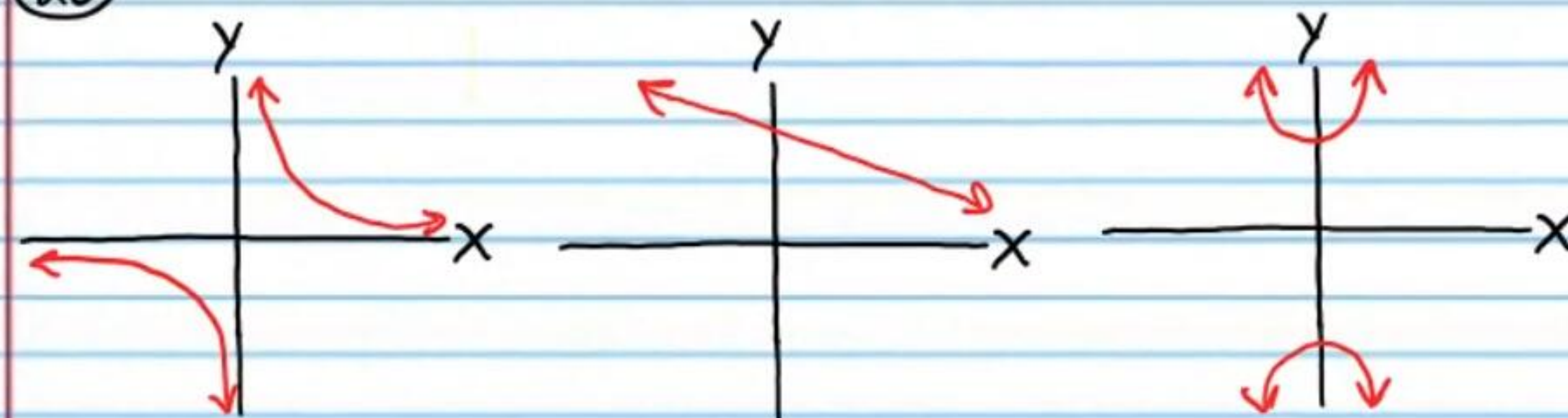
X	Y
16	-8
10	-8
8	-3
0	-1
8	1
0	3

Circle the relations below that cannot be functions.

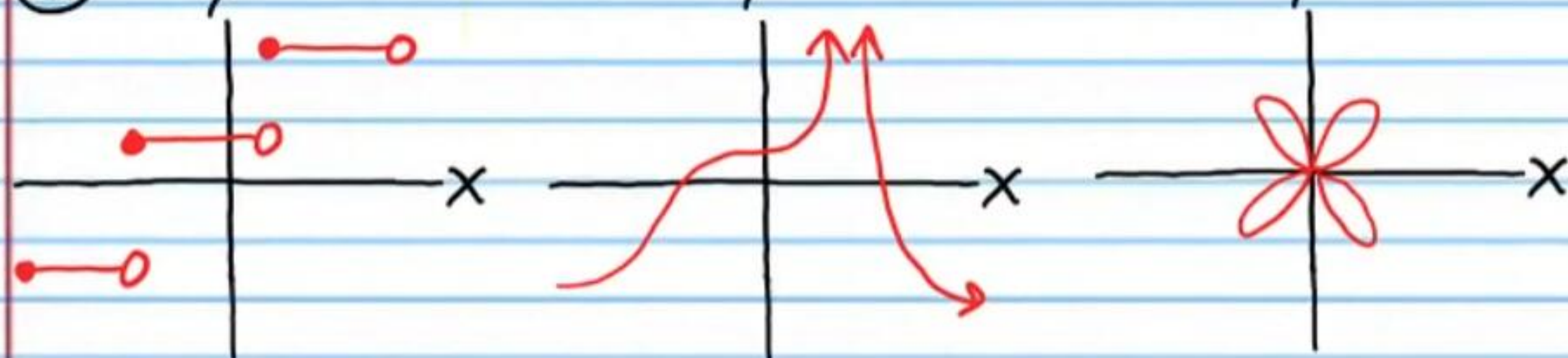
(19)



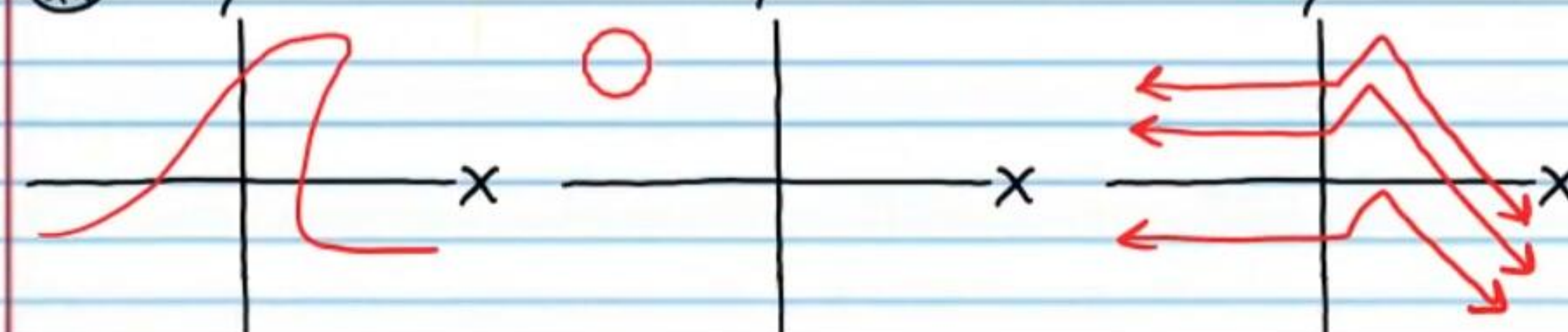
(20)



(21)



(22)



23) If $\frac{0}{x} = 3^{x+4}$, find $\frac{0}{-2}$, $\frac{0}{-1}$, and $\frac{0}{-4}$.

24) If $\boxed{\triangle \triangle \triangle}^b y = b - y$, find $\boxed{\triangle \triangle \triangle}^5 6$, $\boxed{\triangle \triangle \triangle}^7 4$, and $\boxed{\triangle \triangle \triangle}^{-3} -8$.

25) If $f(x) = 3x$, find $f(0)$.

26) If $g(x) = 1 - 2^{2x}$, find $g(1)$.

27) If $h(x) = x^2 - 2x + 5$, find $h(-2)$.

Given the functions below, find the values.

28) $f(x) = \begin{cases} x+4, & 5 < x < 10 \\ \frac{x}{3}, & 10 \leq x \leq 12 \end{cases}$

i) $f(9)$

ii) $f(11)$

iii) $f(12)$

iv) $f(10)$

v) $f(0)$

vi) $f(5)$

29) $f(x) = \begin{cases} 3^{x-1}, & x < 2 \\ -2x, & x \geq 2 \end{cases}$

i) $f(1)$

ii) $f(2)$

iii) $f(0)$

$$\textcircled{30} \quad f(x) = \begin{cases} \frac{3}{x^2}, & x \leq -3 \\ 10 - 4x, & x \geq 0 \end{cases}$$

i) $f(0)$

ii) $f(-3)$

iii) $f(-1)$

iv) $f(-2)$

v) $f(-10)$

vi) $f(1)$

For problem #'s 31-32, find the following:

A. $(f+g)(x)$ B. $(f-g)(x)$

C. $(f \cdot g)(x)$ D. $\left(\frac{f}{g}\right)(x)$

$\textcircled{31} \quad f(x) = 2x - 1$ and $g(x) = x^2$

$\textcircled{32} \quad f(x) = x + x^3$ and $g(x) = \frac{1}{x}$

③③ Find the following values using the functions in problem # 31.

A. $(f \circ g)(x)$

B. $(g \circ f)(x)$

③④ Find the following values using the functions in problem # 32.

A. $(f \circ g)(x)$

B. $(g \circ f)(x)$

Answers

① i) $\boxed{3}$ ii) $\boxed{0}$ iii) $\boxed{-5}$

② i) $\boxed{\sqrt{29}}$ ii) $\boxed{\sqrt{2}}$ iii) $\sqrt{8} = \boxed{2\sqrt{2}}$

③ i) $\boxed{16}$ ii) $\boxed{8}$ iii) $\boxed{1}$

④ i) $\boxed{9}$ ii) $\boxed{3}$ iii) $\boxed{8}$

⑤ i) $\boxed{\frac{3}{16}}$ ii) $\boxed{3}$ iii) $\boxed{\text{undefined}}$

⑥

X	f(x)
0	2
-6	-4
π	$2+\pi$
$\frac{3}{4}$	$2\frac{3}{4}$

⑦

X	9	36	18	-4
f(x)	13	16	$10+3\sqrt{2}$	und

⑧

X	f(x)
0	4
5	$13\frac{1}{2}$

⑨

X	-1	3	2	-10
f(x)	10	22	13	217

⑩ i) $\boxed{\frac{1}{4b}}$ ii) $\boxed{1.25 \text{ or } \frac{5}{4}}$ iii) $\boxed{1}$

⑪ i) $\boxed{-10(a+1)^2 + 5(a+1)}$ ii) $\boxed{-10\pi^2 + 5\pi}$ iii) $\boxed{-10c^2 + 5c}$

⑫ i) $\boxed{-97.5 - \sqrt{9.5}}$ ii) $\boxed{3^{b+1} - 100 - \sqrt{3(3^{b+1}) + 2}}$ iii) $\boxed{-100 - \sqrt{2}}$

⑬ $\square \square \square$

⑭ $\square \square$

⑮ $\square \square$

⑯ $\square \square \square$

⑰ $\square \square$

⑱ $\square \square \square$

⑲ $\square \square \square$

⑳ $\square \square \square$

$$(21) + + (+)$$

$$(22) + + (+)$$

$$(23) \frac{0}{0} = [9]$$

$$\frac{0}{0} = [27]$$

$$\frac{0}{0} = [1]$$

$$(24) \begin{array}{|c|c|c|c|} \hline 5 \\ \hline \end{array} 6 = [-1]$$

$$\begin{array}{|c|c|c|c|} \hline 7 \\ \hline \end{array} 4 = [3]$$

$$\begin{array}{|c|c|c|c|} \hline -3 \\ \hline \end{array} -8 = [5]$$

$$(25) [0]$$

$$(26) [-3]$$

$$(27) [13]$$

$$(28) i) [13]$$

$$ii) [\frac{11}{3}]$$

$$iii) [4]$$

$$iv) [\frac{10}{3}]$$

$$v) \text{Undefined}$$

$$vi) \text{Undefined}$$

$$(29) i) [1]$$

$$ii) [-4]$$

$$iii) [\frac{1}{3}]$$

$$(30) i) [10]$$

$$ii) [\frac{1}{3}]$$

$$iii) \text{Undefined}$$

$$iv) \text{Undefined}$$

$$v) [\frac{3}{100}]$$

$$vi) [6]$$

$$(31) A. [2x-1+x^2] \text{ The order of these terms does not matter.}$$

$$B. [2x-1-x^2]$$

$$C. [(2x-1)(x^2)] \text{ or } [2x^3-x^2]$$

$$D. [\frac{2x-1}{x^2}]$$

$$(32) A. [x+x^3+\frac{1}{x}]$$

$$B. [x+x^3-\frac{1}{x}]$$

$$C. [(x+x^3)(\frac{1}{x})] \text{ or } [1+x^2]$$

$$D. [\frac{x+x^3}{\frac{1}{x}}] \text{ or } [x^2+x^4]$$

$$(33) A. [2x^2-1]$$

$$B. [(2x-1)^2]$$

$$(34) A. [\frac{1}{x} + (\frac{1}{x})^3] \text{ or } [\frac{1}{x} + \frac{1}{x^3}]$$

$$B. [\frac{1}{x+x^3}]$$